

curvaton is in a metastable vacuum, only the temperature corrections can lift the curvaton, therefore we need $V_{\text{th}} > m_0^2 \varphi_{\text{inf}}^2$, so that the thermal effects will overcome the potential barrier. This yields:

$$\alpha_W T^4 > m_0^2 \varphi_{\text{inf}}^2. \quad (455)$$

The thermal mass should be sufficient to trigger curvaton oscillations, otherwise, the curvaton will remain in a metastable vacuum. In order for the flat direction oscillations to start we must have $d^2 V_{\text{th}}/d|\varphi|^2 > H^2$. This leads to:

$$\alpha_W^{1/2} \frac{T^2}{\varphi_{\text{inf}}} > H(T), \quad (456)$$

which always holds in a radiation-dominated phase where $H \simeq T^2/M_{\text{P}}$ (note that $\varphi_{\text{inf}} \ll M_{\text{P}}$). This implies that the $u_1 d_2 d_3$ direction starts oscillating once the condition given in Eq. (455) is satisfied. This happens, when temperature of the Universe is given by

$$T_{\text{osc}} \sim \left(\frac{\varphi_{\text{inf}}}{10^{14} \text{ GeV}} \right)^{1/2} \times 10^9 \text{ GeV}. \quad (457)$$

Although, its a very high temperature, the flat direction curvaton does not evaporate due to the presence of a thermal bath. Thermal scatterings are governed by $\lambda_1^2 \varphi^2 \chi^2$, with φ being the flat direction and, χ collectively denotes the fields in thermal equilibrium. The rate for evaporation is proportional to $\lambda_1^4 \sim (10^{-5})^4$, much feeble to destroy the flat direction by evaporation.

The flat direction curvaton, $u_1 d_2 d_3$, is lifted by $n = 6$ superpotential level, therefore $\varphi_{\text{inf}} \sim 3 \times 10^{14} \text{ GeV}$, see Eq. (451). In order to be a curvaton, the perturbations should match the observed limit, therefore $H_{\text{inf}} \sim 2\pi\varphi_{\text{inf}} \times 10^{-5} \approx 10^{10} \text{ GeV}$.

The final reheat temperature is determined by the radial oscillations of the curvaton. The curvaton oscillations dominate the energy density as the ambient temperature of the plasma redshifts below the T_{osc} very rapidly due to the expansion of the universe. This is the *only* example where the finite temperature effects can render the curvaton oscillations dominating while decaying [136]. The decay of $u_1 d_2 d_3$ curvaton will take place via instant preheating discussed in section V D. The decay happens via SM gauge and Yukawa couplings and the decay products are the MSSM degrees of freedom. There is no residual isocurvature fluctuations and the LSP dark matter can be created from the thermal bath.

Since $u_1 d_2 d_3$ as a curvaton dominates the energy density while decaying, there will be no significant non-Gaussianity. However as we shall show in the next section, there are MSSM